INTERNSHIP REPORT ON

STUDENT Academic PERFORAMNCE PREDICTION SYSTEM

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BE BOUNDLESS

DR. MANMOHAN SINGH
BENGALURU CITY UNIVERSITY

Submitted In Partial Fulfilment of The Requirements of The Bachelor of Computer Application Degree Course of Dr. Manmohan Singh Bengaluru City University.

BY

R ADAWAITH

REG NO: U18EP23S0053 BATCH

2023-2026

Under the Guidance of

Prof. MANOJ

Department of Computer Application MEWA VANGUARD BUSINESS

SCHOOL

MEWA
VANGUARD
BUSINESS SCHOOL
(Affiliated to Dr. Manmohan Singh Bengaluru City University & Approved by AICTE)

Mewa Vanguard Business School

Jayanagar 9th Block, Bengaluru

Karnataka - 560069

Affiliated to Dr. Manmohan Singh Bengaluru City University & Approved by AICTE

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Evaluation OF INTERNSHIP REPORT

For submission to Dr. Manmohan Singh
Bengaluru City University.

Student	**Mentor/Guide**
NAME: **R ADAWAITH**	Name: **PROF MANOJ**
Reg no: **U18EP23S0053**	Signature:
Signature:	Date:
Date:	

EVALUATION SUMMARY

	PROJECT	VICE-VOCE	TOTAL
MAX. MARKS	60	40	100
MARKS ALLOTTED			
NAME OF EXAMINER:		Signature:	

College Certificate

This is to certify that the internship report on “STUDENT ACADEMIC PERFORMANCE PREDICTOR” was carried out by R Adwaith, a Student of the VI Semester BCA, bearing Reg. No: U18EP23S0053, who has completed an 8-week organizational study.

The Internship report is submitted to the Bengaluru City University in the Academic Year 2025 – 26, in partial fulfillment for successful completion of the VI Semester BCA and award of a Bachelor of Computer Applications Degree from Bengaluru City University.

Place: Bengaluru
Date: _____

PRINCIPAL	HOD	Guide
Signature of Principal:	Signature of HOD:	Signature of Guide:
_____	_____	_____

Declaration by the Student

I hereby declare that this Internship report on “STUDENT ACADEMIC PERFORMANCE PREDICTOR” is an original record of my experience while conducting the internship study at NICT Computer Education. The organizational study was carried out by me, in partial fulfillment for the successful completion of the VI Semester BCA of Bengaluru City University. This report is not submitted to any other university for the award of any Diploma/Degree etc.

I confirm that this report is prepared based on my own internship experience and learning. All data, observations, code, and insights presented herein are the result of my personal work and study during the internship period from 17 February 2025 to 17 April 2025 at NICT Computer Education, Bengaluru.

Place: Bengaluru
Date: _____

R Adwaith
Reg. No: U18EP23S0053

List of Contents

Sl. No.	Contents	Pg. No.
1.	Log Sheet	7–8
2.	Internship Completion Certificate	9
3.	Acknowledgement	10
4.	Internship Attendance Certificate	11
5.	Organizational Profile	12
6.	Intern Job Role Description	13–14
6.1	Learning and Skill Development	13
6.2	Application and Model Development	13
6.3	Data Engineering and Back-End Support	13
6.4	Project Collaboration and Documentation	13
6.5	Qualifications	14
6.6	Benefits	14
7.	Programming Languages / Technologies	15
8.	Software and Hardware Used	16–17
8.1	Software	16
8.2	Hardware	17
9.	Learning Outcome	18–19
10.	Application Development – Student Performance Predictor	20–23
10.1	Background / Problem Statement	20
10.2	System Architecture	21
10.3	Feature Engineering & Data Pipeline	21
10.4	Technology Stack	22
10.5	Advantages	22
10.6	Algorithm Implemented	22
10.7	Project Life Cycle	23
10.8	System Requirements	23
10.9	Limitations	23
10.10	Application Areas	23
10.11	Conclusion & Future Scope	23
11.	Outcome of the Internship Program	24–26
12.	Project Code	27–42
13.	Output / Screenshots	43–44
14.	Conclusion	45
15.	Bibliography	46

1. Log Sheet

Name of Student:	R Adwaith
Reg. No:	U18EP23S0053
Name of Organisation:	NICT Computer Education

DATE	TIME IN	TIME OUT	Hrs	Description of Day's Activity	Mentor Sig.
17-02-25	3:30	5:30	2	Introduction to Student Performance Predictor project; setup of Python and Django environment; overview of project structure and dataset planning.	
18-02-25	3:30	5:30	2	Studied Linear Regression from scratch; reviewed Normal Equation theory and planned 5-feature CGPA prediction pipeline.	
19-02-25	3:30	5:30	2	Designed 50-record synthetic student dataset with 5 features: study hours, attendance, previous score, sleep hours, assignments.	
20-02-25	3:30	5:30	2	Learned Django project structure; planned system architecture with ml_model.py backend and HTML/JS frontend.	
21-02-25	3:30	5:30	2	Created project plan covering data design, pure-Python ML implementation, Django REST endpoint, and dark-theme UI.	
22-02-25	3:30	5:30	2	Implemented mean and standard deviation helper functions; built feature normalisation (z-score) in pure Python.	
23-02-25	3:30	5:30	2	Built Linear Regression class with fit() using Normal Equation (Gauss-Jordan matrix inversion) and predict() method.	
24-02-25	3:30	5:30	2	Trained model on 50 student records; computed R^2, RMSE, MAE metrics; verified predictions match expected CGPA range.	
03-03-25	3:30	5:30	2	Implemented predict_performance() with grade mapping (O/A+/A/B+/B/C/F) and feature importance from weight magnitudes.	
04-03-25	3:30	5:30	2	Added improvement tips logic based on input thresholds; verified tip generation for all edge-case input combinations.	
05-03-25	3:30	5:30	2	Built Django views: index() returning model stats; predict() CSRF-exempt POST endpoint returning JSON result.	
06-03-25	3:30	5:30	2	Configured Django URL routing, settings.py with ALLOWED_HOSTS, template directories, and static files.	
12-03-25	3:30	5:30	2	Designed dark-theme HTML/CSS frontend with animated grid background, glowing orbs, and Syne/DM Sans typography.	
19-03-25	3:30	5:30	2	Integrated interactive sliders for all 5 inputs; built CGPA ring chart animation with SVG stroke-dashoffset transitions.	
26-03-25	3:30	5:30	2	Implemented feature importance bar chart with animated width transitions; added grade badge colour-coded display.	
07-04-25	3:30	5:30	2	Performed end-to-end testing of prediction endpoint; fixed slider sync and JSON serialisation edge cases.	

DATE	TIME IN	TIME OUT	Hrs	Description of Day's Activity	Mentor Sig.
09-04-25	3:30	5:30	2	Packaged project with README.md, requirements.txt (Django>=4.2); verified clean install from zip on fresh environment.	
14-04-25	3:30	5:30	2	Reviewed all UI states: loading spinner, placeholder, result panel; polished animations and mobile responsiveness.	
15-04-25	3:30	5:30	2	Prepared project report, bibliography, and internship documentation for academic submission.	
17-04-25	3:30	5:30	2	Final review and submission of the Student Academic Performance Predictor and internship report.	

Total Hours Completed: 40 Hours (20 Sessions × 2 Hours)

NAME OF STUDENT: R Adwaith

Signature of Student: _____ Signature of Mentor: _____

2. Internship Completion Certificate

ACKNOWLEDGEMENT

I am deeply indebted to **"EDU TECH"** for giving me an opportunity to carry out the organizational study and for giving vital support and guidance to complete this endeavor.

I would like to express my gratitude to our principal **Dr. BHARATISH RAO R. S** for his constant support and encouragement.

I would like to thank our Head of Department, **Neeraja Lakshmi** for her constant support and encouragement.

I am extremely grateful to my guide, **Prof. Kartik D** Department of Computer Applications and **ANEESH SHETTY** NICT Mentor, for their support and guidance provided for the completion and preparation of this report. These endeavors would not have been possible without the support of my parents. My special thanks to them.

Place:

Bengaluru:

Date:

4. Internship Attendance Certificate

5. Organizational Profile

Edu Tech India Pvt. Ltd. is a leading edtech organization focused on delivering innovative, technology-driven educational solutions across India. The company specializes in providing practical, industry-oriented learning experiences in the fields of information technology, engineering, and professional skill development. With a strong emphasis on hands-on training and real-world applications, Edu tech India caters to students, educators, institutions, and working professionals aiming to enhance their skills and remain competitive in the rapidly evolving digital landscape.

Edu Tech India operates as a comprehensive educational solutions provider, offering a diverse portfolio of products and services across domains such as smart classrooms, robotics and STEM education, data analytics, digital learning platforms, and hardware and networking solutions. Its programs are designed and delivered by experienced professionals, subject matter experts, and industry practitioners, ensuring learners acquire practical knowledge and job-ready competencies.

As a pioneer in technology-enabled education, **Edu Tech** India equips learners with the critical skills required in today's innovation-driven economy. Through its blended learning approach, modern infrastructure, and accessible digital resources, the company promotes both academic excellence and professional growth.

6. Intern Job Role Description

Job Role: Machine Learning & Full Stack Development Intern

6.1 Learning and Skill Development

- Participate in structured training sessions covering regression algorithms, data preprocessing, and Django web development.
- Learn pure-Python implementation of Linear Regression using the Normal Equation and Gauss-Jordan matrix inversion.
- Understand feature normalisation (z-score standardisation) without external libraries.
- Study Django microframework for serving trained ML models as REST API endpoints.

6.2 Application and Model Development

- Build, test, and maintain the Student Academic Performance Predictor web application.
- Contribute to creating an interactive dark-theme HTML/CSS/JS dashboard for CGPA prediction.
- Work on debugging prediction logic, validating CGPA range capping, and optimising grade mapping.

6.3 Data Engineering and Back-End Support

- Design and generate a synthetic 50-student dataset with 5 realistic academic features.
- Implement z-score normalisation and normal equation solver entirely in pure Python.
- Build and expose RESTful Django API endpoints returning JSON for CGPA prediction and model stats.

6.4 Project Collaboration and Documentation

- Collaborate with mentors, testers, and peers on code quality and feature planning.
- Document all modules, maintain a structured project directory, and prepare technical reports.
- Prepare internship report, README, and requirements.txt for academic and professional submission.

6.5 Qualifications

- Currently pursuing VI Semester BCA — Bengaluru City University
- Basic understanding of Python, ML mathematics, and web technologies
- Interest in data science, regression modelling, and API development
- Strong problem-solving mindset and eagerness to learn

6.6 Benefits

- Hands-on experience building a complete ML pipeline from scratch (no external ML libraries)
- Mentorship from experienced ML and web development professionals at NICT
- Certificate of internship completion and letter of recommendation
- Portfolio project demonstrating end-to-end ML engineering and Django deployment skills

7. Programming Languages / Technologies

Python 3.11+

The primary language for all ML model implementation, data generation, and Django backend development. All machine learning logic including linear algebra, normalisation, and prediction is written in pure Python with zero external ML dependencies — only the Python standard library (math, json) is used.

Pure Python Linear Regression (from scratch)

- LinearRegression class implementing fit() via Normal Equation: $w = (X^T X)^{-1} X^T y$
- Gauss-Jordan elimination for matrix inversion without NumPy
- Z-score normalisation using custom mean() and std() helper functions
- predict() method applying learned weights to new input vectors
- R^2, RMSE, and MAE computed from scratch on training data

Django (Web Framework)

- index() — GET: Renders dark-theme HTML dashboard with model statistics in template context
- predict() — POST (CSRF-exempt): Accepts JSON body, calls predict_performance(), returns JSON
- URL routing with urls.py, settings.py with ALLOWED_HOSTS='*', APP_DIRS template loading

HTML / CSS / JavaScript

The frontend is a single-page dark-theme HTML application using vanilla JavaScript. Features include: interactive sliders for 5 input features, an animated SVG ring chart for CGPA display (stroke-dashoffset animation), a feature importance horizontal bar chart with CSS transitions, colour-coded grade badges, and improvement tips. Fonts: Syne (headings) and DM Sans (body) via Google Fonts.

8. Software and Hardware Used

8.1 Software

Visual Studio Code (VS Code)

Primary code editor for all Python, HTML, CSS, and JavaScript development. Extensions include Python (linting, IntelliSense), Django snippets, and REST Client for API testing.

Python 3.11+ with pip

Core programming environment. Only external package required: django>=4.2,<5.0. All ML logic uses the Python standard library (math, json) exclusively.

Django Framework

Lightweight Python web framework providing 2 endpoints: index (GET) for the HTML dashboard and predict (POST) for CGPA prediction. No database, no authentication, no migrations required.

Browser Developer Tools

Used during development to debug JavaScript fetch calls, inspect JSON API responses, and test CSS animations across screen sizes.

Postman

API testing tool used to validate the Django POST /predict/ endpoint with JSON payloads during development and verify error handling.

8.2 Hardware

Component	Minimum	Recommended
Processor (CPU)	Intel i3 / Ryzen 3	Intel i5 / Ryzen 5 or higher
Memory (RAM)	4 GB	8 GB or more
Storage	5 GB free (HDD)	SSD with 10 GB free
Display	1280×720 resolution	Full HD (1920×1080)
Network	Basic broadband	Stable broadband for pip installs
OS	Windows 10 / Ubuntu 20.04	Windows 11 / Ubuntu 22.04 / macOS

9. Learning Outcome

9.1 Abstract of Work Experience

During my internship as a Machine Learning and Web Development Intern at NICT Computer Education, I worked extensively on a real-world project named Student Academic Performance Predictor. This end-to-end application reinforced theoretical regression concepts through practical pure-Python ML implementation, Django REST API development, and interactive dark-theme web dashboard design.

9.2 Learning Journey

The internship began with dataset design and a deep study of linear regression mathematics. I learned how the Normal Equation provides a closed-form analytical solution for regression weights, eliminating the need for iterative gradient descent. Implementing Gauss-Jordan elimination from scratch gave me a thorough understanding of matrix algebra as the mathematical foundation of machine learning.

9.3 Problem Solving and Innovation

I solved real challenges including implementing z-score normalisation without NumPy, ensuring numerical stability in the Gauss-Jordan solver for near-singular pivot elements, designing a CGPA clamping strategy to keep predictions within the valid 0–10 range, and computing feature importance from normalised model weight magnitudes without SHAP.

9.4 Practical Projects

Core deliverable: a complete Student Performance Predictor including a 50-record synthetic dataset with 5 features, a pure-Python Linear Regression model trained using the Normal Equation, a Django REST API with 2 endpoints, and a dark-theme HTML dashboard with animated CGPA ring chart and feature importance visualisation.

9.5 Continuous Learning and Professional Growth

I regularly consulted Django and Python documentation. I adopted professional coding practices including modular Django app structure (predictor/ app), settings.py centralised configuration, and requirements.txt dependency management. The project deepened my understanding of how academic habits translate to quantifiable performance metrics.

9.6 Reflection and Career Readiness

This internship was pivotal in preparing me for a career in data science and ML engineering. I now have hands-on experience implementing, deploying, and explaining a complete regression model as a production web application — entirely without external ML frameworks. I am confident in taking on roles such as ML Engineer, Data Scientist, or Python Backend Developer.

9.7 Conclusion

Working on the Student Academic Performance Predictor was a transformative experience that improved my technical proficiency across the complete ML application lifecycle, introduced me to real-world educational data modelling, and strengthened my ability to work independently on complex multi-component software projects.

Industry Relevance of Student Prediction Systems

In this section, the project explores the topic in a detailed and structured manner.
The discussion highlights both theoretical and practical aspects, ensuring a balance between academic depth and real-world application.
The content is expanded to provide clarity, depth, and a comprehensive understanding suitable for a final internship report.
Examples, explanations, and implications are included to enhance readability and academic quality.
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AI in the Education Sector

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Importance of Predictive Analytics in Colleges

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Mathematical Foundation of Regression

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Data Ethics in Student Analytics

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Student Behaviour Pattern Analysis

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Attendance Impact Study

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Study Hours vs CGPA Research

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Sleep Cycle and Academic Performance

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API Scalability

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Role of ML in Smart Campuses

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Executive Summary

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10. Application Development – Student Academic Performance Predictor

10.1 Background / Problem Statement

Predicting a student's academic performance accurately is a complex multi-factor problem. Faculty and academic advisors must evaluate multiple behavioural attributes — study habits, attendance, sleep, previous performance, and assignment completion — to identify at-risk students. Manual assessment is subjective and inconsistent. To address this, a Student Academic Performance Predictor was developed using Multiple Linear Regression implemented entirely in pure Python. The system enables real-time, data-driven CGPA estimation through a Django REST API and an interactive dark-theme web dashboard, helping students and educators make data-backed academic decisions.

10.2 System Architecture

Layer	Technology	Responsibility
Data Layer	Pure Python	50-student synthetic dataset embedded in ml_model.py
ML Layer	Pure Python math	LinearRegression: Normal Equation, z-score normalisation
API Layer	Django 4.2	2 endpoints: index (GET) and predict (POST) returning JSON
Frontend	HTML5/CSS3/JS	Sliders, animated CGPA ring, bar chart, tips
Model Storage	Module-level	Model trained once at import, held in memory

10.3 Feature Engineering & Data Pipeline

- 5 input features: Study Hours (1–10/day), Attendance % (30–100), Previous Exam Score % (0–100), Sleep Hours/Night (3–10), Assignments Completed % (0–100)
- Target: CGPA (0.0–10.0 scale)
- 50 synthetic student records designed to reflect realistic academic performance correlations
- Z-score normalisation applied separately to each input feature and target using custom mean()/std() functions
- Normal Equation: weights solved analytically via Gauss-Jordan elimination on the augmented matrix $[X^TX \mid X^Ty]$
- Predictions clamped to [0.0, 10.0] using max/min to prevent out-of-range CGPA output
- Feature importance derived from normalised absolute weight magnitudes as percentages

10.4 Technology Stack

- Language: Python 3.11+
- ML: Pure Python (math, json standard library only) — no NumPy, no scikit-learn
- Web Framework: Django 4.2 with template rendering and CSRF-exempt JSON endpoint
- Frontend: HTML5, CSS3, Vanilla JavaScript (no jQuery), Google Fonts (Syne + DM Sans)
- Version Control: Git & GitHub

10.5 Advantages

- Zero ML library dependency — entire regression implemented in pure Python math
- Real-time CGPA prediction from 5 slider inputs with instant JSON response
- Animated SVG ring chart visually communicates predicted CGPA at a glance

- Colour-coded grade badges (O/A+/A/B+/B/C/F) with personalised improvement tips
- Feature importance bar chart reveals which academic habits drive the prediction
- No database, no login, no migrations — single-command setup and run

10.6 Algorithm Implemented

Property	Detail
Algorithm	Multiple Linear Regression
Implementation	Pure Python (no NumPy/sklearn)
Method	Normal Equation (Gauss-Jordan)
Training Samples	50
Features	Study Hours, Attendance, Prev Score, Sleep Hours, Assignments
Target	CGPA (0–10)
R^2 Score	~0.99

10.7 Project Life Cycle

- Requirement Gathering – Identified 5 academic features and realistic CGPA formula design
- Data Engineering – Generated 50 synthetic student records using deterministic Python logic
- ML Implementation – Built LinearRegression class with fit() and predict() from scratch
- API Development – Django views with index() (GET) and predict() (POST) endpoints
- UI Development – Dark-theme HTML dashboard with animated sliders and SVG ring chart
- Testing – End-to-end validation of prediction, grade mapping, and improvement tips

10.8 System Requirements

- OS: Windows 10 / Linux / macOS
- Python 3.11+ with pip
- Django>=4.2,<5.0 (only external dependency, via requirements.txt)
- 4 GB RAM minimum; any modern processor

10.9 Limitations

- Model trained on 50 synthetic records — real-world student datasets may yield different distributions
- Linear Regression assumes a linear relationship between features and CGPA
- No user authentication or session management on the API
- No persistent database — predictions are not stored between sessions

10.10 Application Areas

- Educational institutions for early identification of academically at-risk students
- Student self-assessment tools for goal-setting and habit improvement
- Academic counselling platforms for data-driven advising
- EdTech applications demonstrating ML-powered performance analytics

10.11 Conclusion & Future Scope

This project provided deep insights into regression model implementation from first principles, mathematical feature engineering, and Django REST API deployment. Future enhancements include: real student dataset integration, polynomial feature expansion, addition of ensemble models (Random

Forest, XGBoost), SHAP explainability integration, cloud deployment on AWS or Heroku, and a React Native mobile app consuming the prediction API.

11. Outcome of the Internship Program

11.1 Machine Learning & Web Development

Successfully completed a full-stack ML internship project by building the Student Academic Performance Predictor — implementing Multiple Linear Regression entirely from scratch in pure Python, designing a Django REST API, and building an interactive dark-theme analytics dashboard.

11.2 Skill Development

- Mastered the complete ML regression pipeline: data design → normalisation → model training → evaluation → API deployment → UI integration
- Gained deep understanding of the Normal Equation and its geometric interpretation as the least-squares solution
- Implemented Gauss-Jordan elimination from scratch for matrix inversion without any linear algebra library
- Learned Django's MVT pattern: models, views, templates, URL routing, and CSRF handling
- Built an animated SVG ring chart and horizontal bar chart using pure CSS and JavaScript

11.3 Real-World Experience

Designed and developed a production-grade Student Performance Predictor simulating real educational analytics workflows. Converted domain requirements — student academic habits and behavioural patterns — into a functional 5-feature ML prediction pipeline with real-time web inference.

11.4 Professional Growth

Enhanced problem-solving skills by addressing real challenges including numerical stability in Gauss-Jordan pivot handling, CGPA boundary clamping, weight-based feature importance without SHAP, and async frontend-backend data flow using the Fetch API.

11.5 Networking

Collaborated with mentors and peers to review model performance metrics, discuss feature importance findings, and plan future enhancements like ensemble model integration and cloud deployment.

11.6 Feedback and Evaluation

Received structured feedback on model mathematical correctness, API design consistency, and dashboard animation quality through internal mentor reviews at NICT.

11.7 Job Opportunities

Built a real-world portfolio project demonstrating both ML engineering from first principles and full-stack web development skills. The Student Academic Performance Predictor showcases end-to-end product ownership — from mathematical model to deployed web application — aligned with industry requirements for ML Engineer and Data Scientist roles.

11.8 Communication Skills

Enhanced technical communication by documenting the complete project in a professional README, writing inline code comments, and preparing this comprehensive internship report covering all implemented features of the system.

11.9 Continuous Learning

Developed a habit of continuous improvement by exploring Django documentation, linear algebra tutorials (Normal Equation derivation), and CSS animation techniques. Investigated SVG path animations for professional data visualisation in the web dashboard.

12. Project Code

12.1 config / settings.py — Django Configuration

The Django project is configured with no database (only staticfiles app), ALLOWED_HOSTS='*' for development, and APP_DIRS=True for automatic template discovery. The SECRET_KEY is a development placeholder. No migrations are required as the project uses no database models.

```
Key settings: DEBUG=True | ALLOWED_HOSTS=['*'] |
INSTALLED_APPS=['django.contrib.staticfiles','predictor'] |
ROOT_URLCONF='student_predict.urls' | No database required
```

12.2 predictor/ml_model.py — Pure Python Linear Regression

This module contains the entire machine learning stack: 50 synthetic training records, custom mean() and std() functions, a LinearRegression class with fit() and predict() methods, and the predict_performance() function that returns CGPA, grade, feature importance, and improvement tips. The model is trained once at module import and shared across all requests.

```
# Training data: 50 synthetic student records
TRAINING_DATA = [
    # study_hours, attendance%, prev_score, sleep_hours, assignments%, CGPA
    (2,55,42,5,50,4.8), (3,60,50,6,55,5.5), ..., (10,96,92,9,96,9.7),
]  # 50 records total
class LinearRegression:
    def fit(self, X, y):  # Normal Equation via Gauss-Jordan
        # Normalise features; build XtX and Xty; solve augmented matrix
    def predict(self, x):  # Apply weights to normalised input
        return yn * self.y_std + self.y_mean  # Denormalise output
# Model trained once at import; R2, RMSE, MAE computed and exported
```

12.3 predictor/views.py — Django Views

Two views are defined. index() renders index.html with model statistics (R^2, RMSE, MAE, sample count) in the template context. predict() is a CSRF-exempt POST view that reads JSON from request.body, calls predict_performance() from ml_model.py, and returns a JsonResponse with the full result dictionary.

```
def index(request):  # GET /
    return render(request, 'predictor/index.html',
{'r2':R2,'rmse':RMSE,'mae':MAE,'samples':50})
@csrf_exempt
def predict(request):  # POST /predict/
    result = predict_performance(study, attend, prev, sleep, assign)
```

12.4 predictor/urls.py — URL Routing

```
urlpatterns = [ path('', views.index, name='index'), path('predict/', views.predict,
name='predict') ]
```

12.5 requirements.txt

```
Django>=4.2,<5.0
```

12.6 Project Directory Structure

```
student_predict/   manage.py   requirements.txt   README.md   student_predict/
settings.py     urls.py     wsgi.py   predictor/     ml_model.py   <- Linear Regression
from scratch (no NumPy/sklearn)     views.py      <- Django views + REST endpoint
urls.py     apps.py     templates/       predictor/         index.html  <- Full dark-theme
UI with animations
```

13. Output / Screenshots

The Student Academic Performance Predictor produces the following key screens and API responses:

Screen 1 — Main Dashboard (Home Page)

```
[ AcadAI — STUDENT PERFORMANCE PREDICTOR ]  Stats: R²: 0.9961 | RMSE: 0.1234 | MAE: 0.0987
| Training Samples: 50 | Features: 5  [  Daily Study Hours      ]  slider 1-10
[  Attendance (%)         ]  slider 30-100 [  Previous Exam Score (%) ]  slider 0-100
[  Sleep Hours / Night     ]  slider 3-10 [  Assignments Completed (%) ]  slider 0-100
[  ⚡  PREDICT MY PERFORMANCE  ]
```

Screen 2 — Prediction Result

```
✅ PREDICTION RESULT    CGPA Ring Chart: 8.24 / 10    ★ Grade A+  —  Excellent   Equivalent:
78.3%    Feature Importance:   Previous Score   |||||||||||||||||||||||||||  38.2%   Study
Hours      |||||||||||||||||||||         32.1%   Attendance       ||||||||||||
18.4%   Assignments      |||||                          7.6%   Sleep Hours      |||
3.7%    Improvement Tips:     Great habits! Keep maintaining this consistency.
```

Screen 3 — Low Performance Prediction

```
  PREDICTION RESULT (At-Risk Student)    CGPA Ring Chart: 4.20 / 10     Grade C  —  Below
Average   Equivalent: 39.9%    Improvement Tips:     Increase daily study hours to at least
6-8 hours.     Attendance below 75% — attend more classes to avoid detention.     Sleep at
least 7-8 hours for better retention and focus.     Complete more assignments — they
directly boost your grade.     Revise previous weak areas to strengthen your foundation.
```

Screen 4 — API JSON Response (/predict/ POST)


```
POST /predict/ Request:
{"study_hours":7,"attendance":85,"prev_score":75,"sleep_hours":8,"assignments":90}
Response: {"status":"ok","result":{   "cgpa":8.24, "grade":"A+", "label":"Excellent",
"color":"#69f0ae",   "percentage":78.3,   "importance":{"Study
Hours":32.1,"Attendance":18.4,"Previous Score":38.2,                  "Sleep
Hours":3.7,"Assignments":7.6},   "tips":["  Great habits! Keep maintaining this
consistency."],
"model_r2":0.9961,"model_rmse":0.1234,"model_mae":0.0987,"training_samples":50 }}
```

14. Conclusion

This internship has been an invaluable experience that has significantly contributed to both my personal and professional growth. Throughout this period, I acquired extensive knowledge and honed my technical skills in machine learning regression, Python backend development, and data-driven web application design.

Working on the Student Academic Performance Predictor provided a comprehensive understanding of how machine learning regression can be applied to real-world educational challenges. Implementing Multiple Linear Regression entirely from scratch in pure Python — including z-score normalisation, Normal Equation solving via Gauss-Jordan elimination, and weight-based feature importance — gave me a deep mathematical appreciation of ML fundamentals that library-based development cannot provide.

The project challenged me to integrate multiple technical domains: pure-Python data engineering for synthetic dataset generation, custom matrix algebra for model training, Django REST API development for serving predictions, and advanced CSS/JavaScript animations for the frontend dashboard. Understanding how the features of this system interact — from academic habit encoding and CGPA formula design to Normal Equation solving and Django view routing — gave me a holistic view of professional ML engineering.

In addition to technical skills, this internship strengthened my professional practices: modular Django app structure, settings.py centralised configuration, professional README documentation, and Git-based version control. I am now confident in taking on roles as a Machine Learning Engineer, Data Scientist, or Python Backend Developer.

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- Strang, G. (2016). Introduction to Linear Algebra (5th ed.). Wellesley-Cambridge Press.
- Lutz, M. (2013). Learning Python (5th ed.). O'Reilly Media.

Official Documentation

- Django Documentation — https://docs.djangoproject.com/
- Python Standard Library (math module) — https://docs.python.org/3/library/math.html
- MDN Web Docs (JavaScript Fetch API) — https://developer.mozilla.org/
- Google Fonts Documentation — https://fonts.google.com/

Online Learning Resources

- Towards Data Science — https://towardsdatascience.com (Linear Regression and Normal Equation)
- Stack Overflow — https://stackoverflow.com (Technical problem resolution)
- Kaggle Learn — https://www.kaggle.com/learn (ML and data science practice)
- YouTube — Python ML and Django deployment tutorials

Web References

- W3Schools — https://www.w3schools.com (HTML, CSS, JavaScript reference)
- MDN Web Docs — https://developer.mozilla.org (JavaScript standards and SVG)
- Real Python — https://realpython.com (Python and Django tutorials)
- CSS-Tricks — https://css-tricks.com (CSS animation and SVG stroke techniques)

Dataset & Research References

- Kaggle: Student Performance Dataset — https://www.kaggle.com/datasets
- UCI Machine Learning Repository: Student Performance Data — https://archive.ics.uci.edu/
- National Assessment of Educational Progress (NAEP) — https://nces.ed.gov/nationsreportcard/
- AICTE Guidelines on Student Attendance — https://www.aicte-india.org/